Target-Effect Residual Covariance for REPAIR

REPAIR Development Memo (GPT 5.6 Luna)

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1 Purpose

The original REPAIR formulation considers a two-level mixed model with independent residuals,

$$\mathbf{y}_j = \mathbf{X}_j\boldsymbol{\beta} + \mathbf{Z}_j\mathbf{b}_j + \mathbf{e}_j, \qquad \mathbf{e}_j \sim N(\mathbf{0}, \mathbf{R}_j),$$

where the usual special case is

$$\mathbf{R}_j = \sigma^2\mathbf{I}.$$

For models with multiple random-effects levels, REPAIR must specify which random effect is the target and which random effects are treated as part of its prediction error. This memo formulates the corresponding target-specific covariance matrix.

The central principle is:

> A target random effect is predicted using generalized least squares, while all non-target random effects are incorporated into the residual covariance.

2 Partitioning the mixed model

Suppose the fitted model contains several random-effects blocks. Partition them into one target block and one or more nuisance blocks:

$$\mathbf{y}_j = \mathbf{X}_j\boldsymbol{\beta} + \mathbf{Z}_{T,j}\mathbf{b}_{T,j} + \sum_{g\neq T} \mathbf{Z}_{g,j}\mathbf{b}_{g,j} + \mathbf{e}_j.$$

Here:

- j indexes the target cluster;
- $\mathbf{b}_{T,j}$ is the target random-effects vector;
- $\mathbf{Z}_{T,j}$ is its design matrix;
- $\mathbf{b}_{g,j}$ is a non-target random-effects vector;
- $\mathbf{Z}_{g,j}$ is its design matrix;
- $\mathbf{e}_j$ is the ordinary residual vector.

Assume that

$$\mathrm{Var}(\mathbf{b}_{g,j}) = \mathbf{G}_g, \qquad \mathrm{Var}(\mathbf{e}_j) = \mathbf{R}_{e,j}.$$

The standard mixed-model assumption is that the target random effects, non-target random effects, and ordinary residuals are mutually independent.

3 Target and nuisance decomposition

Define the nuisance component as

$$\mathbf{u}_{T,j} = \sum_{g\neq T} \mathbf{Z}_{g,j}\mathbf{b}_{g,j} + \mathbf{e}_j.$$

The model can then be written as

$$\mathbf{y}_j = \mathbf{X}_j\boldsymbol{\beta} + \mathbf{Z}_{T,j}\mathbf{b}_{T,j} + \mathbf{u}_{T,j}.$$

The covariance of this nuisance component is

$$\mathbf{R}_{T,j} = \mathrm{Var}(\mathbf{u}_{T,j}).$$

Under independence across random-effects blocks,

$$\boxed{\mathbf{R}_{T,j} = \mathbf{R}_{e,j} + \sum_{g\neq T} \mathbf{Z}_{g,j}\mathbf{G}_g\mathbf{Z}_{g,j}^{\mathsf{T}}}$$

This is the covariance matrix used for target-effect GLS prediction.

The target random effect itself is not included in $\mathbf{R}_{T,j}$. Including it would treat the quantity being predicted as part of its own measurement error.

4 Example: person and day effects

Consider the three-level model:

```
y ~ 1 + x + (1 + x | person) + (1 | person:day)
```

For person j, write the model as

$$\mathbf{y}_j = \mathbf{X}_j\boldsymbol{\beta} + \mathbf{Z}_{P,j}\mathbf{b}_{P,j} + \mathbf{Z}_{D,j}\mathbf{b}_{D,j} + \mathbf{e}_j.$$

Here:

- $\mathbf{b}_{P,j}$ contains the person-level random intercept and slope;
- $\mathbf{b}_{D,j}$ contains the day-level random effects;
- $\mathbf{e}_j$ is the ordinary residual.

If the person effect is the target, the day effects are nuisance effects. The target-specific covariance is

$$\boxed{\mathbf{R}_{P,j} = \mathbf{R}_{e,j} + \mathbf{Z}_{D,j}\mathbf{G}_D\mathbf{Z}_{D,j}^{\mathsf{T}}}$$

where $\mathbf{G}_D$ is the covariance matrix of the day-level random effects.

If the ordinary residuals are independent and homoskedastic,

$$\mathbf{R}_{e,j} = \sigma^2\mathbf{I}.$$

If the residuals are heteroskedastic or correlated, $\mathbf{R}_{e,j}$ is the fitted residual covariance matrix from the model.

For a random-intercept day effect, $\mathbf{Z}_{D,j}$ is an indicator matrix. Observations collected on the same day therefore receive a covariance contribution from the shared day effect.

5 Target-effect GLS prediction

After estimating the fixed effects, define the fixed-effect-adjusted response:

$$\mathbf{r}_j = \mathbf{y}_j - \mathbf{X}_j\widehat{\boldsymbol{\beta}}.$$

If the model includes an offset $\mathbf{o}_j$, use

$$\mathbf{r}_j = \mathbf{y}_j - \mathbf{o}_j - \mathbf{X}_j\widehat{\boldsymbol{\beta}}.$$

The target-effect likelihood-only GLS estimate is

$$\boxed{\widetilde{\mathbf{b}}_{T,j} = \left(\mathbf{Z}_{T,j}^{\mathsf{T}}\mathbf{R}_{T,j}^{-1}\mathbf{Z}_{T,j}\right)^{-1}\mathbf{Z}_{T,j}^{\mathsf{T}}\mathbf{R}_{T,j}^{-1}\mathbf{r}_j}$$

The corresponding prediction-error covariance is

$$\boxed{\boldsymbol{\Psi}_{T,j} = \left(\mathbf{Z}_{T,j}^{\mathsf{T}}\mathbf{R}_{T,j}^{-1}\mathbf{Z}_{T,j}\right)^{-1}}$$

Treating the fixed-effects and variance-component estimates as plug-in quantities gives the additive-error representation

$$\widetilde{\mathbf{b}}_{T,j} = \mathbf{b}_{T,j} + \boldsymbol{\varepsilon}_{T,j},$$

with

$$\mathrm{Var}(\boldsymbol{\varepsilon}_{T,j}) = \boldsymbol{\Psi}_{T,j}.$$

The matrices $\boldsymbol{\Psi}_{T,j}$ are the measurement-error covariance matrices supplied to the second-stage M-MOM estimator.

6 Why the target block is excluded

The full marginal covariance of the response is

$$\mathbf{V}_j = \mathbf{R}_{e,j} + \sum_g \mathbf{Z}_{g,j}\mathbf{G}_g\mathbf{Z}_{g,j}^{\mathsf{T}}.$$

This includes the target contribution

$$\mathbf{Z}_{T,j}\mathbf{G}_T\mathbf{Z}_{T,j}^{\mathsf{T}}.$$

Using $\mathbf{V}_j$ in the target GLS calculation would treat variation due to $\mathbf{b}_{T,j}$ as measurement error rather than as the signal being estimated.

Therefore, the covariance used for target prediction must exclude the target block:

$$\mathbf{R}_{T,j} = \mathbf{V}_j - \mathbf{Z}_{T,j}\mathbf{G}_T\mathbf{Z}_{T,j}^{\mathsf{T}}.$$

Equivalently,

$$\mathbf{R}_{T,j} = \mathbf{R}_{e,j} + \sum_{g \neq T} \mathbf{Z}_{g,j}\mathbf{G}_g\mathbf{Z}_{g,j}^{\mathsf{T}}.$$

7 Multiple non-target blocks

If there are several non-target random-effects blocks, their covariance contributions are additive:

$$\mathbf{R}_{T,j} = \mathbf{R}_{e,j} + \mathbf{Z}_{1,j}\mathbf{G}_1\mathbf{Z}_{1,j}^{\mathsf{T}} + \mathbf{Z}_{2,j}\mathbf{G}_2\mathbf{Z}_{2,j}^{\mathsf{T}} + \cdots.$$

In code, the construction is conceptually:

```
R_target <- R_error

for (group in nuisance_groups) {
  Z_group <- random_design[[group]][rows, , drop = FALSE]
  G_group <- random_covariance[[group]]

  R_target <- R_target +
    Z_group %*% G_group %*% t(Z_group)
}
```

The target design is extracted separately:

```
Z_target <- target_design[
  rows,
  target_columns,
  drop = FALSE
]
```

8 Correlated target and nuisance effects

The additive formula assumes that target and nuisance random effects are uncorrelated:

$$\mathrm{Cov}\left(\mathbf{b}_{T,j}, \mathbf{b}_{g,j}\right) = \mathbf{0}.$$

If target and nuisance random effects are correlated, then the nuisance component is not independent of the target. In that case, the conditional mean and covariance require additional cross-covariance terms, and the simple additive-error representation may not hold without modification.

The current REPAIR extension should therefore state the following working assumption:

> Target and non-target random-effects blocks are mutually independent, consistent with the block-diagonal random-effects covariance structures used by the supported mixed-model backends.

9 Relation to two-level REPAIR

The original two-level formulation is recovered when there are no non-target random-effects blocks:

$$\sum_{g \neq T} \mathbf{Z}_{g,j} \mathbf{G}_g \mathbf{Z}_{g,j}^{\mathsf{T}} = \mathbf{0}.$$

Then

$$\mathbf{R}_{T,j} = \mathbf{R}_{e,j}.$$

With independent homoskedastic residuals,

$$\mathbf{R}_{T,j} = \sigma^2 \mathbf{I},$$

and therefore

$$\mathbf{\Psi}_{T,j} = \sigma^2 \left(\mathbf{Z}_{T,j}^{\mathsf{T}} \mathbf{Z}_{T,j}\right)^{-1}.$$

Thus, the original two-level equations are a special case of the target-specific formulation.

10 Implications for `lme4` and `nlme`

The model-extraction layer should store:

```
ordinary residual covariance
all random-effects design matrices
all random-effects covariance matrices
grouping variables and row mappings
```

The scoring layer should construct the target-specific covariance after the target group is known:

```
R_target =
    R_error
    + covariance contributions from all non-target random effects
```

For `lme4`, the ordinary residual covariance is typically

$$\mathbf{R}_{e,j} = \operatorname{diag}\left(\frac{\widehat{\sigma}^2}{w_i}\right),$$

where w_i are fitted prior weights.

For `nlme`, $\mathbf{R}_{e,j}$ may include:

- the fitted residual scale;
- correlation structures such as `corAR1`;
- variance functions such as `varIdent` or `varPower`.

Once these backend-specific components have been normalized, the target-specific GLS calculation is the same.

11 Scope and caveats

This extension relies on the following working assumptions:

1. The fixed-effects coefficients and variance components are treated as plug-in estimates.
2. Target and non-target random-effects blocks are mutually independent.
3. The target design matrix has sufficient within-cluster rank.
4. The target-specific residual covariance is positive definite.
5. Correlated target random-effect components are estimated jointly.
6. Non-target random-effects design matrices are aligned with the observations in each target cluster.
7. The fitted model and extracted covariance components correspond to the same effective observations.

The final condition is especially important for nested structures. When targeting person effects, for example, day-level design matrices must be restricted to the observations belonging to each person before computing

$$\mathbf{Z}_{D,j}\mathbf{G}_D\mathbf{Z}_{D,j}^{\mathsf{T}}.$$